NIHAL POTHUNOORI

804-549-9190 | nihal4@illinois.edu | https://nihalpothunoori.github.io/ | US Citizen

EDUCATION

University of Illinois at Urbana-Champaign Expected May 2027
B.S. in Computer Engineering

Relevant Coursework: Control Systems; Deep Learning; Principles of Safe Autonomy; Analog Signal Processing; Computer Systems & Programming

SKILLS

Programming: C++, Python (PyTorch, OpenCV, NumPy), CUDA, Matlab, Java, C#/.NET

ML & Robotics: ROS2, Deep/Reinforcement Learning, Diffusion Models, Vision-Language-Action Models, Computer Vision, State Estimation, Simulation (Gazebo, Isaac Sim/Lab)

Tools: Linux, Git, Docker; Design: CAD (Fusion360), KiCAD, Altium, PCB Design (I2C, SPI)

EXPERIENCE

Rivian Jun 2026 – Present
Machine Learning Intern (Incoming) *Champaign, IL*

- Optimizing world models and vision-language-action models to run on constrained custom hardware (RAP1).

DASLABS, University of Illinois Jan 2026 – Present
Undergraduate Researcher *Champaign, IL*

- Designing a VLA architecture in Python/PyTorch for a hybrid rigid-soft arm that perceives environmental geometry to identify viable anchor and bracing structures for the soft link; optimizing manipulation efficiency beyond passive obstacle avoidance.
- Built a depth-camera pipeline to isolate the deformable soft-arm backbone (not possible in simulation); used real-time point-cloud processing and SAM3-based binary mask segmentation for online policy evaluation against ground-truth pose benchmarks.

College of Education, University of Illinois Aug 2025 – Present
Web Developer/IT Support *Champaign, IL*

- Built an automated enterprise-level permissions management system for ∼2,500 users via a high-performance .NET 8 C# library integrated with Azure Active Directory to streamline identity and access management across the department.
- Developing an intuitive web-based interface in HTML to enable non-technical administrators to execute complex operations on user roles, reducing manual IT support tickets and ensuring 99.9% consistency in organizational permission hierarchies.

Autonomous Materials Systems Group, University of Illinois Aug 2025 – Jan 2026
Undergraduate Researcher *Champaign, IL*

- Designed a 3-axis robotic manipulator using inverse kinematics and automated pipette dispenser for the preparation of DCPD polymer samples within a DARPA-funded laboratory focused on rapid, autonomous material synthesis and chemical discovery.
- Architected and deployed a PyTorch-based reinforcement learning framework for closed-loop process control, optimizing between environmental variables and mechanical actuation to achieve a 75% reduction in total sample preparation time.

Toast Gastropub Jun 2025 – Aug 2025
Line Cook *Richmond, VA*

PROJECTS

Soccer Counterfactual Video Generation Apr 2026

- Implemented a text-conditioned video generation model from scratch in Python/PyTorch inspired by the Wan paper; architected a full VAE + Diffusion Transformer (DiT) pipeline capable of generating alternate soccer realities from natural language prompts, with frames decoded simultaneously from a learned continuous latent space.
- Built text conditioning via a T5 encoder into DiT cross-attention (K/V); applied AdaLN timestep embeddings and classifier-free guidance for controllable inference; trained end-to-end on ∼20,000 5-second clips from SoccerNetv2 (HuggingFace).

Laundry Robot Jan 2026

- Developed a ROS2 mobile platform to autonomously retrieve scattered laundry; uses a YOLOv8 segmentation model for clothing detection; perception utilizes a vision-to-world service to map 2D image pixels into 3D coordinates in lieu of an RGB-depth camera.
- Implemented an Extended Kalman Filter (EKF) to fuse asynchronous odometry and visual data for localization and state estimation; estimates informed a Stanley Controller that minimizes cross-track error to provide precise path-following controls.

GNN School Zone Detection – Hackathon Apr 2026

- Architected a Graph Neural Network to identify school zones from sparse map data; model inherently reasons over geographic structure that flat neural networks cannot express; outperformed all other hackathon teams in accuracy metrics.
- Implemented and trained a Graph Neural Network from scratch in PyTorch on Iowa school zone data (speed limits, road signs, landmarks, school locations); the GNN predicts speed limits for all surrounding road segments given a known school zone anchor, achieving 99.7% accuracy within +/-5 mph and 86.7% within +/-2 mph.

LEADERSHIP

Hardware Lead Jan – May 2025
iRobotics Micromouse, University of Illinois *Champaign, IL*

- Led hardware design for maze-solving robots; ran weekly design reviews and taught PCB/PCBA fundamentals (15+ members).
- Fabricated custom PCBs in KiCAD with I2C/SPI sensor interfaces and analog filtering to ensure signal integrity for data acquisition.